

42.510 Statistical Learning for Data Science

Logistic regression

Bikramjit Das

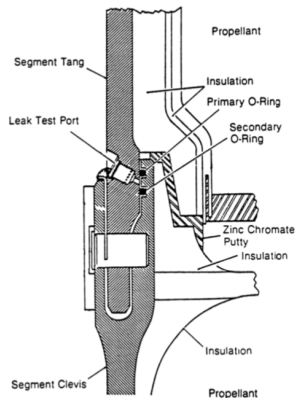
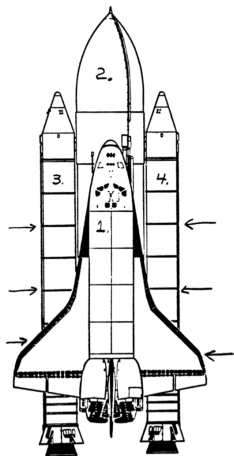
Week 3, Lecture 2

The Challenger space shuttle

- January 28, 1986.
- 7 crew members.
- Included a teacher (**Teacher in space** program).
- One objective: observe Halley's comet.

If you are interested in this topic, check out: *Challenger: The Final Flight* – Netflix

Challenger schematic and the O-Ring



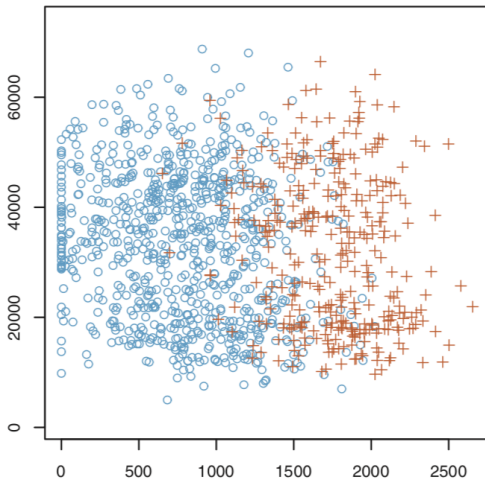
Left: Schematic of Challenger shuttle: subsystem 1 houses crew & system control, subsystem 2 is the external liquid-fuel tank, and subsystems 3 & 4 are the solid rocket motors (Morton Thiokol).

Right: Solid Rocket Motor Subsection (Morton Thiokol).

Challenger disaster: Rogers commission report excerpt

The report of the fateful series of NASA and Morton Thiokol meetings, telephone conferences, notes and facsimile transmissions on January 27, the night before the launch of 51-L, shows that only limited consideration was given to the past history of O-ring damage in terms of temperature. The managers compared as a function of temperature, the flights for which thermal distress of O-rings had been observed - not the frequency of occurrences based on all flights. In such a comparison, there is nothing irregular in the distribution of O-ring distress over the spectrum of joint temperatures at launch between 53° and 75° Fahrenheit.

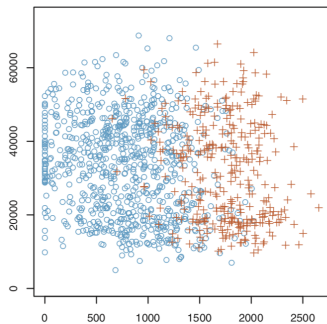
Classification



Classification

- Objective: Predict binary/categorical variables with some respondent, explanatory variables.
- O-Ring failure $\sim$ Temperature, Pressure
- Many methods: Linear discriminant analysis, perceptrons, even linear regression.
- We concentrate on Logistic Regression for this class.
- Next class: Discriminant Analysis.
- *Caveat:* LDA is a **generative** method, whereas logistic regression is a **discriminative** method.

Classification



Problem setup

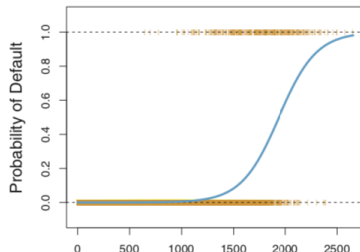
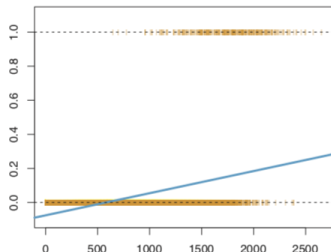
- n = Number of observations
- p = Number of predictor variables
- $y \in \{0, 1\}$ = Dependent variable (binary)
- $x_1, \dots, x_p$ = Independent variables (predictors)

We want to predict y with a function $f(\mathbf{x})$.

- Model: $y^* = \beta_0 + \beta_1 x_1 + \dots + \beta_p x_p$ where

$$y = \begin{cases} 1 & \text{if } y^* > 0, \\ 0 & \text{if } y^* \leq 0. \end{cases}$$

Linear regression vs. Logistic regression



- Linear regression: we may end up predicting numbers beyond $[0, 1]$.
- We can check that the solution given by linear regression for binary classification is equivalent to LDA.

Logistic regression

- Define $\beta^\top = (\beta_0, \dots, \beta_p)$ and $\mathbf{x} = (1, x_1, \dots, x_p)^\top$.
- Logistic regression model

$$\begin{aligned}\mathbb{P}(Y = 1|\mathbf{x}, \beta) &=: p(\mathbf{x}, \beta) = \frac{\exp(\beta^\top \mathbf{x})}{1 + \exp(\beta^\top \mathbf{x})} \\ &= \frac{\exp(\beta_0 + \beta_1 x_1 + \dots + \beta_p x_p)}{1 + \exp(\beta_0 + \beta_1 x_1 + \dots + \beta_p x_p)}, \\ \mathbb{P}(Y = 0|\mathbf{x}, \beta) &= 1 - p(\mathbf{x}, \beta) = \frac{1}{1 + \exp(\beta^\top \mathbf{x})}.\end{aligned}$$

- We define **Odds ratio** as

$$\text{Odds} = \frac{\mathbb{P}(Y = 1|\mathbf{x}, \beta)}{\mathbb{P}(Y = 0|\mathbf{x}, \beta)} = e^{\beta^\top \mathbf{x}}.$$

- Therefore: $\log(\text{Odds}) = \beta^\top \mathbf{x}$ which is linear.

Logistic regression

- In linear regression we have y being predicted by $\beta^\top \mathbf{x}$.
 - ▶ Hence, if $x_j \uparrow$ by 1 unit, then $y \uparrow$ by β_j units.
- In logistic regression, we have $\log(\text{Odds}) = \beta^\top \mathbf{x}$.
 - ▶ Let $\beta_j > 0$: If $x_j \uparrow$ by 1 unit, then $\log(\text{Odds}) \uparrow$ by β_j units.
 - ▶ Let $\beta_j < 0$: If $x_j \uparrow$ by 1 unit, then $\log(\text{Odds}) \downarrow$ by $|\beta_j|$ units.
 - ▶ In fact, we may think in either case Odds is multiplied by e^{β_j} .

Logistic regression: estimation of parameters

- Data: $(y_i, \mathbf{x}_i), i = 1, \dots, n$ where $\mathbf{x}_i^\top = (1, x_{i1}, \dots, x_{ip})$.
- First we write down the likelihood function of the data:

$$\begin{aligned} L(\beta) &= \prod_{i=1}^n \mathbb{P}(Y = 1 | \mathbf{x}_i, \beta)^{y_i} \mathbb{P}(Y = 0 | \mathbf{x}_i, \beta)^{1-y_i} \\ &= \prod_{i=1}^n p(\mathbf{x}_i, \beta)^{y_i} (1 - p(\mathbf{x}_i, \beta))^{1-y_i}. \end{aligned}$$

- We want to find $\max_{\beta} L(\beta)$ or equivalently $\max_{\beta} \ln L(\beta)$.

$$\begin{aligned} LL(\beta) &= \ln L(\beta) = \sum_{i=1}^n y_i \ln p(\mathbf{x}_i, \beta) + \sum_{i=1}^n (1 - y_i) \ln (1 - p(\mathbf{x}_i, \beta)) \\ &= \sum_{i=1}^n \left[y_i \beta^\top \mathbf{x}_i - \ln(1 + e^{\beta^\top \mathbf{x}_i}) \right]. \end{aligned}$$

- Although it is not analytically possible to find the minima of this function, since the function is concave, a local minima is the global minima.
- Hence we may use any numerical method to find the minima, e.g., Newton-Raphson.

Quality of fit: deviance and AIC

- How do we know which parameter gives a good model?
- A measure like R^2 or adjusted- R^2 does not make sense for logistic regression.
- **Null deviance**: here we find logistic regression with the intercept term (equivalent to SST).

- ▶ Find β_0 such that $\mathbb{P}(\widehat{Y} = 1) = \bar{y} = \frac{e^{\hat{\beta}_0}}{1 + e^{\hat{\beta}_0}}$.

- ▶ Then Null deviance $= -2LL(\hat{\beta}_0)$.

- **Residual deviance** $= -2LL(\hat{\beta})$.

- ▶ Clearly residual deviance is lower than null deviance and keeps getting lower as we increase the number of dimensions p .

- **Akaike's information criterion** (AIC): This modifies residual deviance so that we penalize for the number of variables used.

$$\text{AIC} = -2LL(\hat{\beta}) + 2(p + 1).$$

Quality of fit: confusion matrix

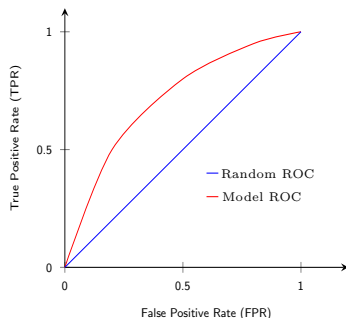
- Predict based on a threshold t and the estimated $\hat{\beta}$.

$$\begin{aligned}\mathbb{P}(Y_i = 1 | \mathbf{x}_i, \hat{\beta}) \geq t, & \quad \text{predict } \hat{Y}_i = 1, \\ < t, & \quad \text{predict } \hat{Y}_i = 0.\end{aligned}$$

	Actual = 0	Actual = 1
Predict = 0	True Negative (TN)	False Negative (FN)
Predict = 1	False Positive (FP)	True Positive (TP)

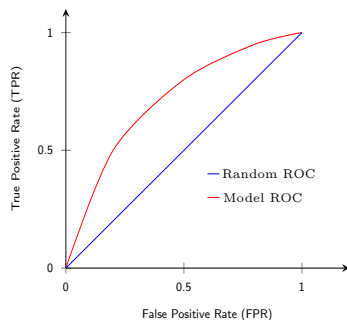
- True negative rate: $\text{TNR} = \frac{TN}{FP+TN}$ (Specificity),
- False positive rate: $\text{FPR} = \frac{FP}{FP+TN}$ (Type I error),
- True positive rate: $\text{TPR} = \frac{TP}{FN+TP}$ (Sensitivity),
- False negative rate: $\text{FNR} = \frac{FN}{FN+TP}$ (Type II error),
- Overall Accuracy: $\text{Accuracy} = \frac{TN+TP}{FP+TN+FN+TP}$.

Quality of fit: Receiver operating characteristic (ROC) curve



- Previously we fixed t and found the **confusion matrix**.
- **ROC**: Plot $(FPR(t), TPR(t))$ for $0 \leq t \leq 1$.
- $t = 0$: By definition, $TN = FN = 0$ and hence $FPR = TPR = 1$.
- $t = 1$: By definition, $FPR = TPR = 0$.
- Often a **baseline** model uses $t = 0.5$.

Quality of fit: Area under the curve (AUC)



- Area under the curve: **AUC**.
- **AUC** for the random ROC = 0.5.

$\text{AUC} = \mathbb{P}(\text{a randomly chosen point with } Y = 1 \text{ is ranked higher than a randomly chosen point with } Y = 0).$

References

- Dalal et al. (1989), *Risk Analysis of the Space Shuttle: Pre-Challenger Prediction of Failure*, Journal of Statistical Association, Vol. 84, No. 408, pp. 945-957.
- Hastie et al. (2001), *The Elements of Statistical Learning*, Chapter 4.
- James et al. (2023), *Introduction to Statistical Learning*, Chapter 4.